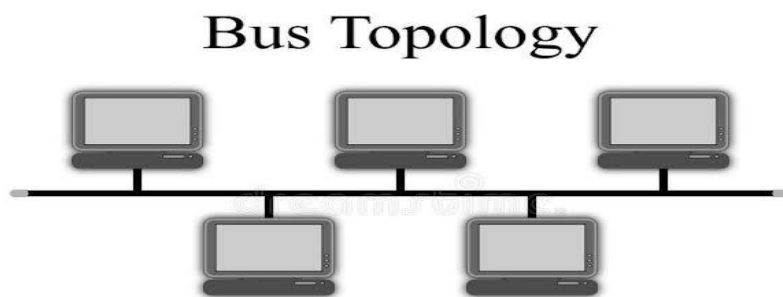


Network Topologies

- 1) Draw and label diagrams of star, bus, and mesh network topologies on paper or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 lines.

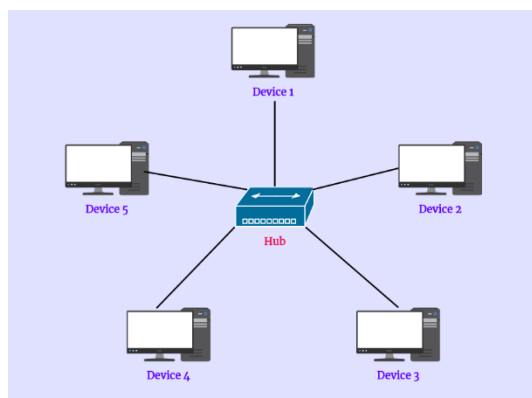
- **BUS TOPOLOGY**



Example : **Small Office File Sharing**

All computers share one communication cable, making it simple and inexpensive for small networks.

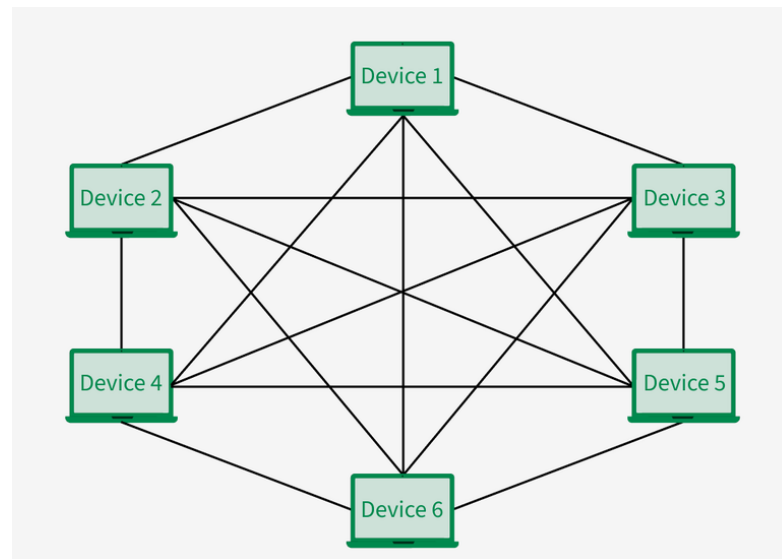
- **STAR TOPOLOGY**



Example : **INSTAGRAM**

Instagram users connect to central servers. If one user's connection fails, others continue working normally

- **MESH TOPOLOGY**



Example : **WhatsApp**

Messages travel through multiple servers and routes, ensuring reliable communication even if one route fails.

- 1) Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for 'Best Use Case' and give an example from apps you use (e.g., Instagram, Zomato, Paytm).

Topology	Pros	Cons	Best Use Case	Example App
Star	Easy to manage, Easy to add devices, Fast performance, One device failure doesn't affect others	Switch failure stops network, More cable required, Higher cost, Central device required	Schools, Offices, Homes	Instagram
Bus	Low cost, Easy installation, Less cable needed, Good for small networks	Cable failure stops network, Difficult troubleshooting, Slow with many devices, Limited length	Small offices	Paytm
Ring	Equal access, No data collision, Predictable performance, Good under heavy load	One device failure affects network, Difficult expansion, Complex maintenance, Higher downtime	Token Ring Networks	Banking Systems
Mesh	Very reliable, Multiple paths, High security, No single point of failure	Very expensive, Lots of cables, Complex setup, Difficult maintenance	Military, Data Centers	WhatsApp

- 2) Imagine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.

- **Star Topology**

A Star topology is best because all players connect to a central game server. It provides fast communication, low latency, and easy troubleshooting. If one player's connection is lost, other players continue playing without interruption. It is also easy to expand by adding more players or servers.

- 3) Given this scenario: A small Flipkart-style warehouse wants to connect 8 computers for order processing and inventory updates. Recommend a network topology, sketch a quick diagram, and explain your reasoning in 2-3 lines.
Hint: Consider factors like cost, ease of troubleshooting, and future expansion.

- **Star Topology**